Level 4 · about 30 h · Python 3.8+

Project: An address card kept in a dict and saved as a text file
Examples run on CPython 3.12.3. **You will be able to:**

- Store several values in a list and read one by index
- Look up a value in a dict by its key
- Tell a set apart from a list
- Read and write a small text file with pathlib

Lists, Dicts and Files

A list keeps an order

A list is a mutable sequence. Indexes start at 0. append adds at the end. The Python wiki's TimeComplexity page describes CPython list costs; it is community notes, not a language guarantee.

```
names = ["Ada", "Grace"]
names.append("Guido")
print(names[0], len(names))
print(names[1:3])
```

```
Output (CPython 3.12.3):
Ada 3
['Grace', 'Guido']
```

A dict looks up by key

A dict maps each key to one value. `in` tests keys, not values.

```
card = {"name": "Ada", "year": 1815}
print(card["name"])
print("year" in card)
print(card.get("city", "unknown"))
```

```
Output (CPython 3.12.3):
Ada
```

```
True
unknown
```

A set keeps unique items

A set does not keep order you can rely on, and it drops duplicates.

```
tools = {"pen", "ink", "pen"}
print(len(tools))
print("ink" in tools)
```

```
Output (CPython 3.12.3):
2
True
```

A file should say its encoding

`pathlib.Path` is the object-oriented way to name a file. `write_text` and `read_text` take an encoding. This sample uses a temporary directory so the build does not leave a file behind.

```
import tempfile
from pathlib import Path
folder = Path(tempfile.mkdtemp())
path = folder / "card.txt"
path.write_text("Ada\n", encoding="utf-8")
print(path.read_text(encoding="utf-8").strip())
```

```
Output (CPython 3.12.3):
Ada
```

Check yourself

1. What is the index of the first item in a list?
2. Does `in` on a dict test keys or values?
3. Why pass `encoding="utf-8"` when you read text?

Sources

- Python docs: Built-in types

- Python docs: pathlib
- Python tutorial: Input and output
- Python wiki: TimeComplexity

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